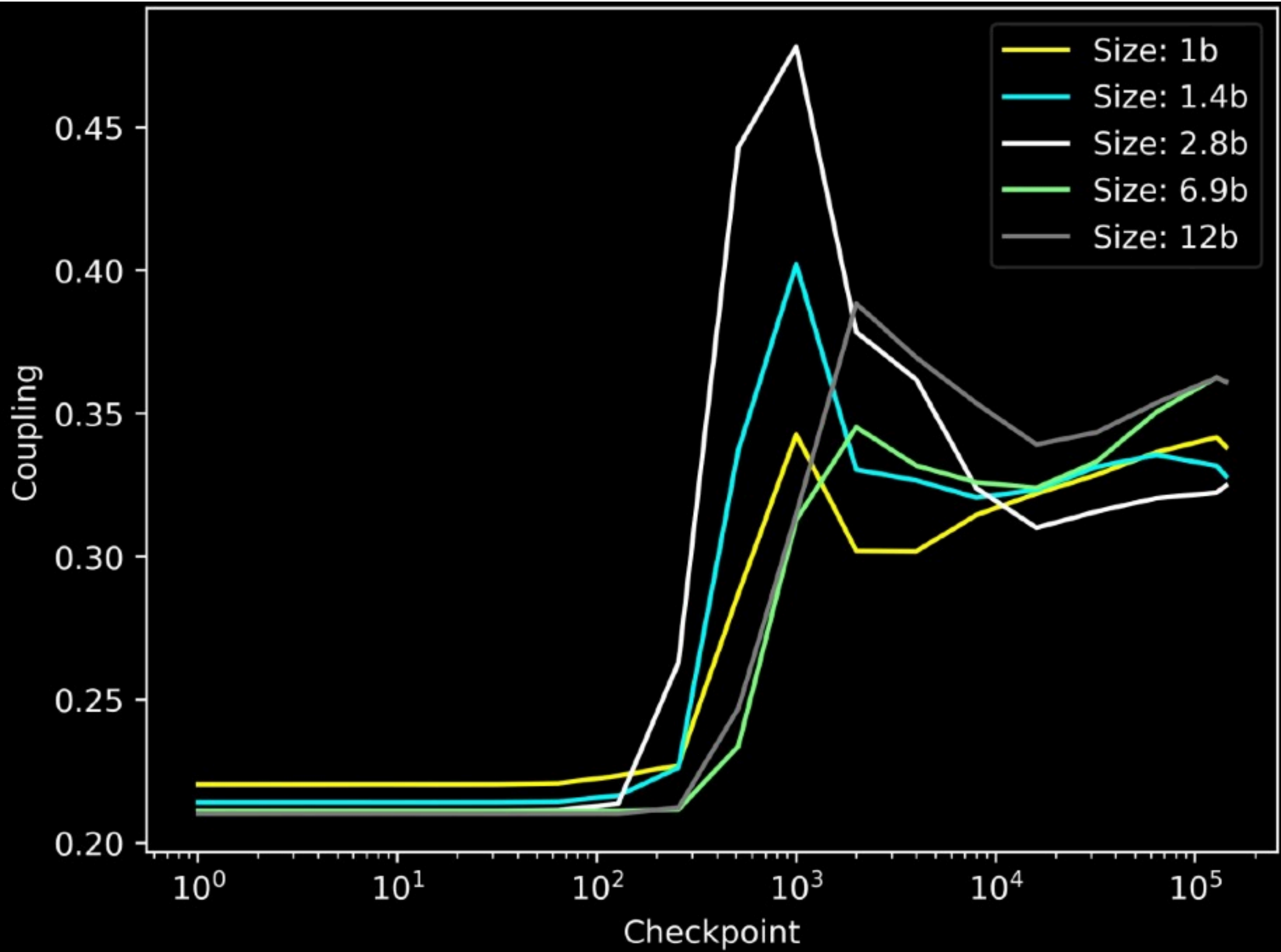
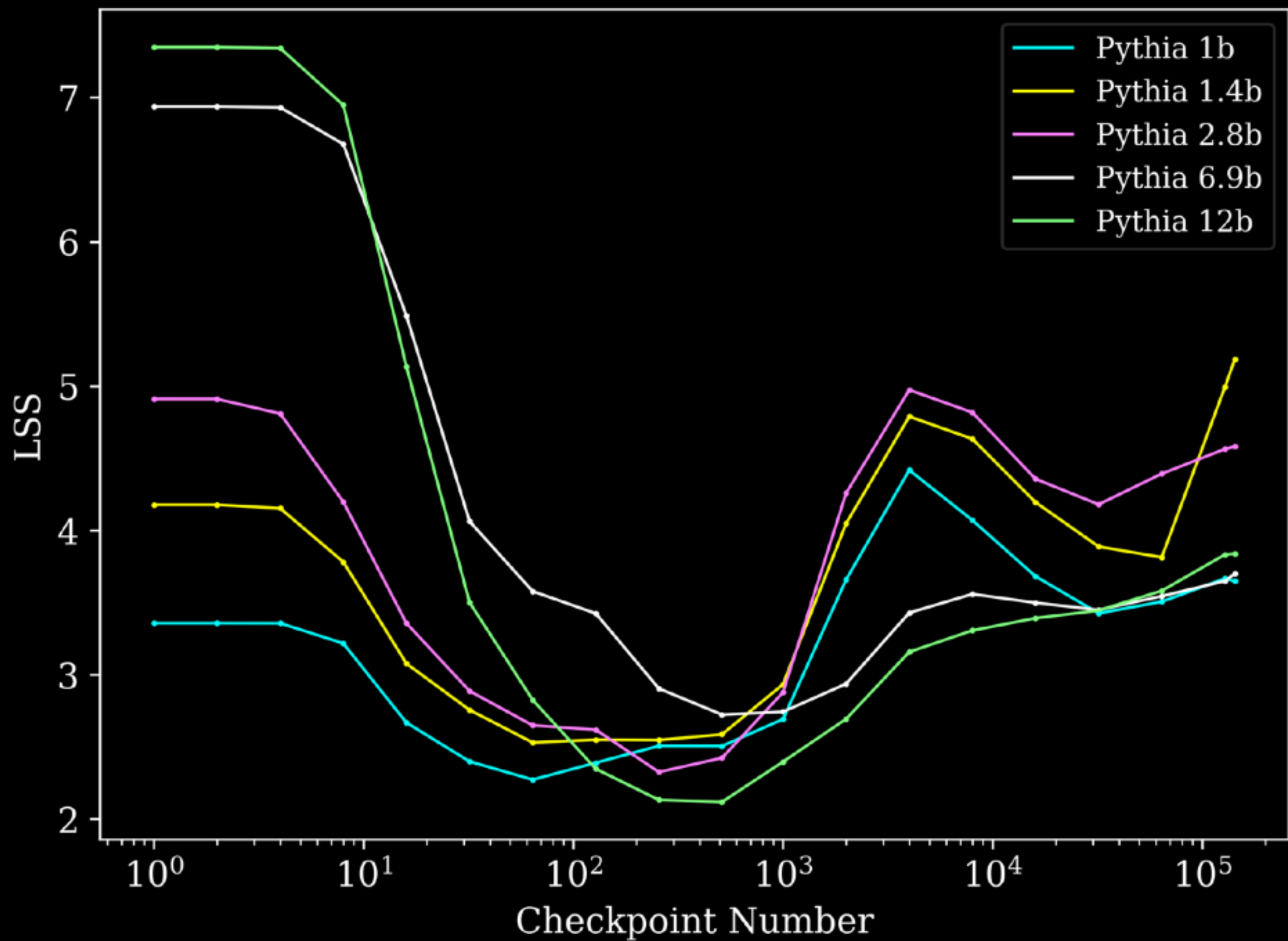
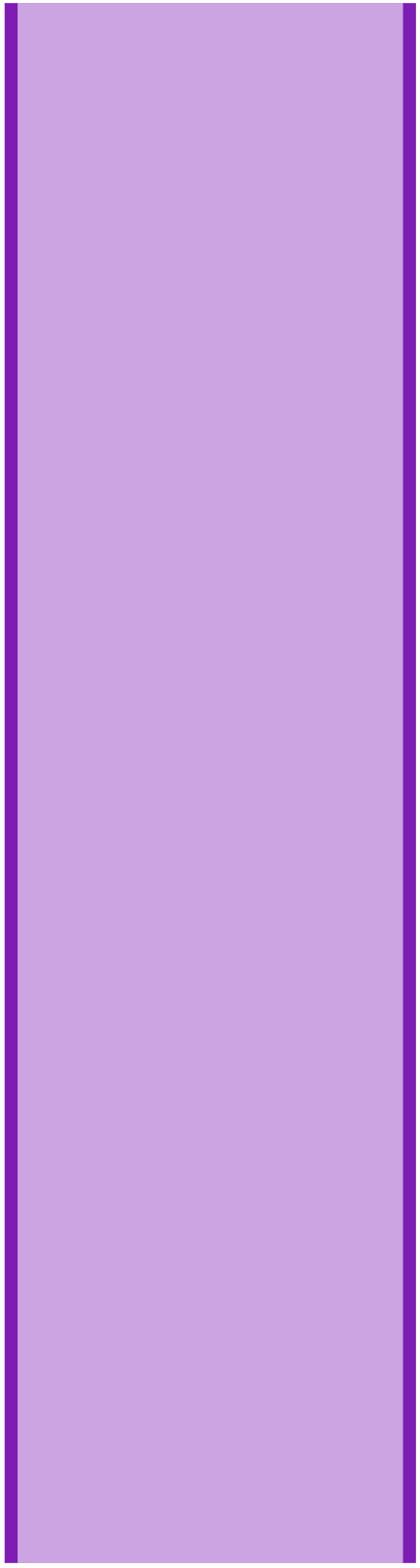
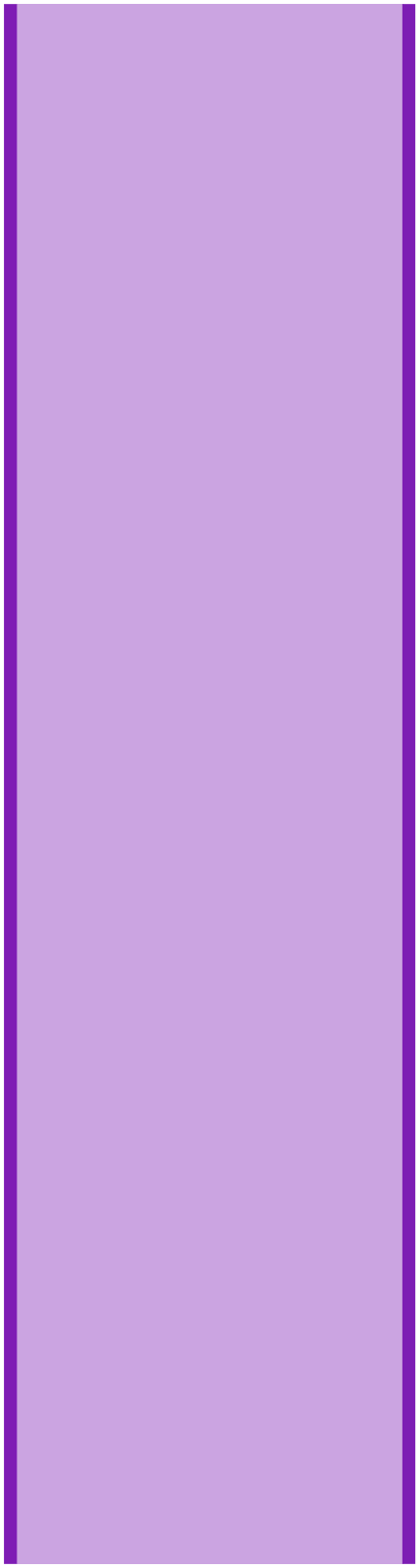




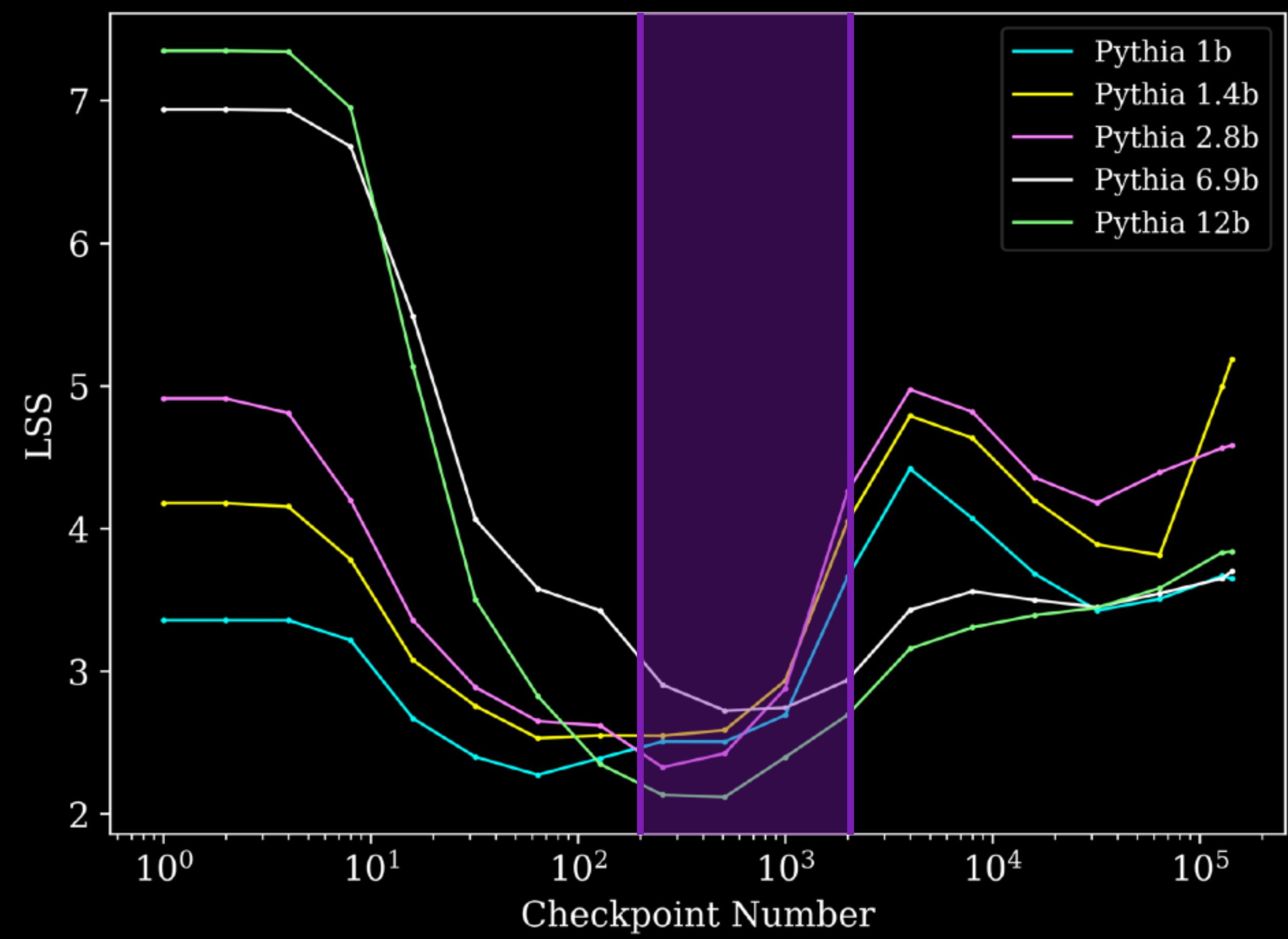
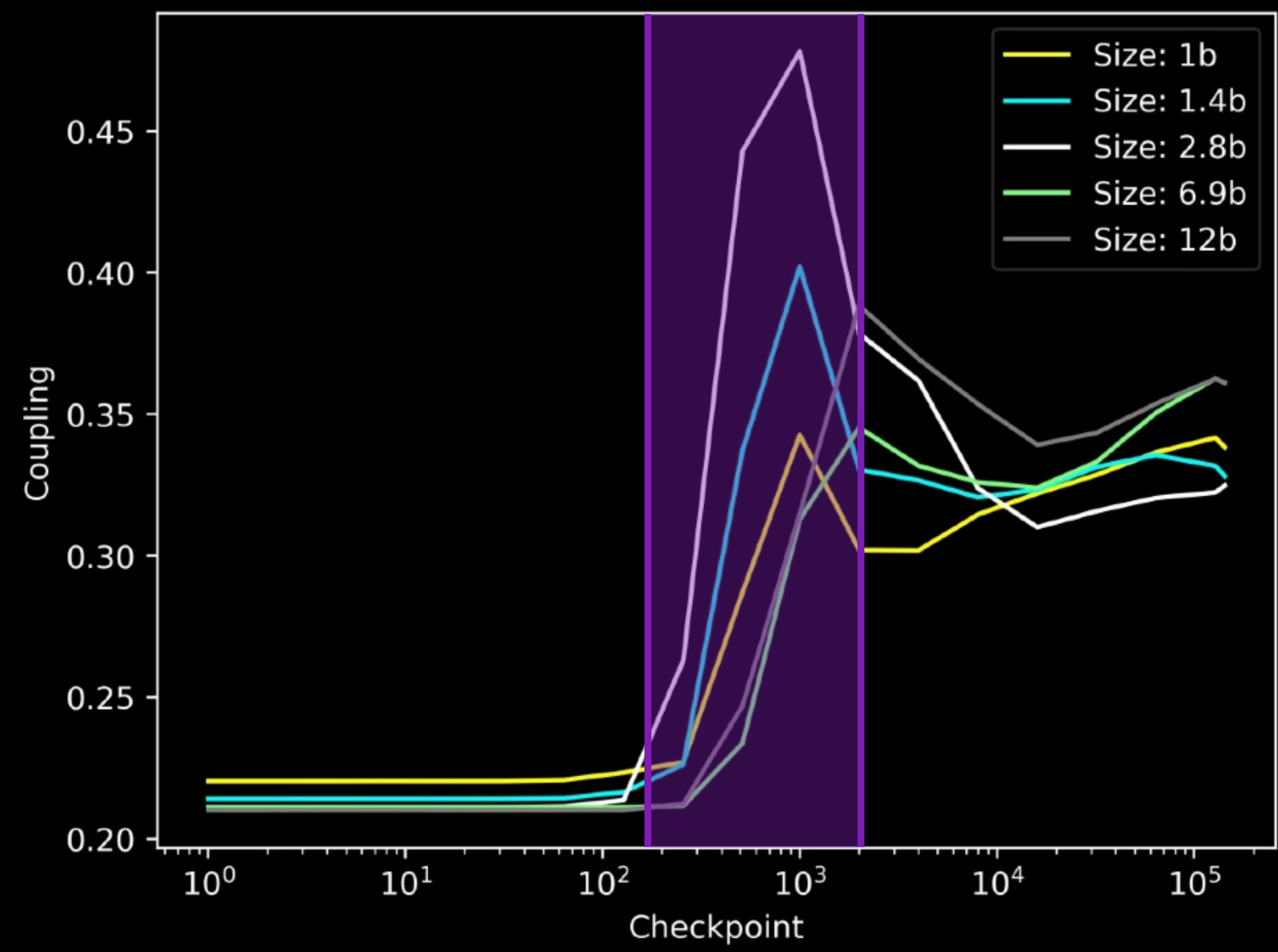






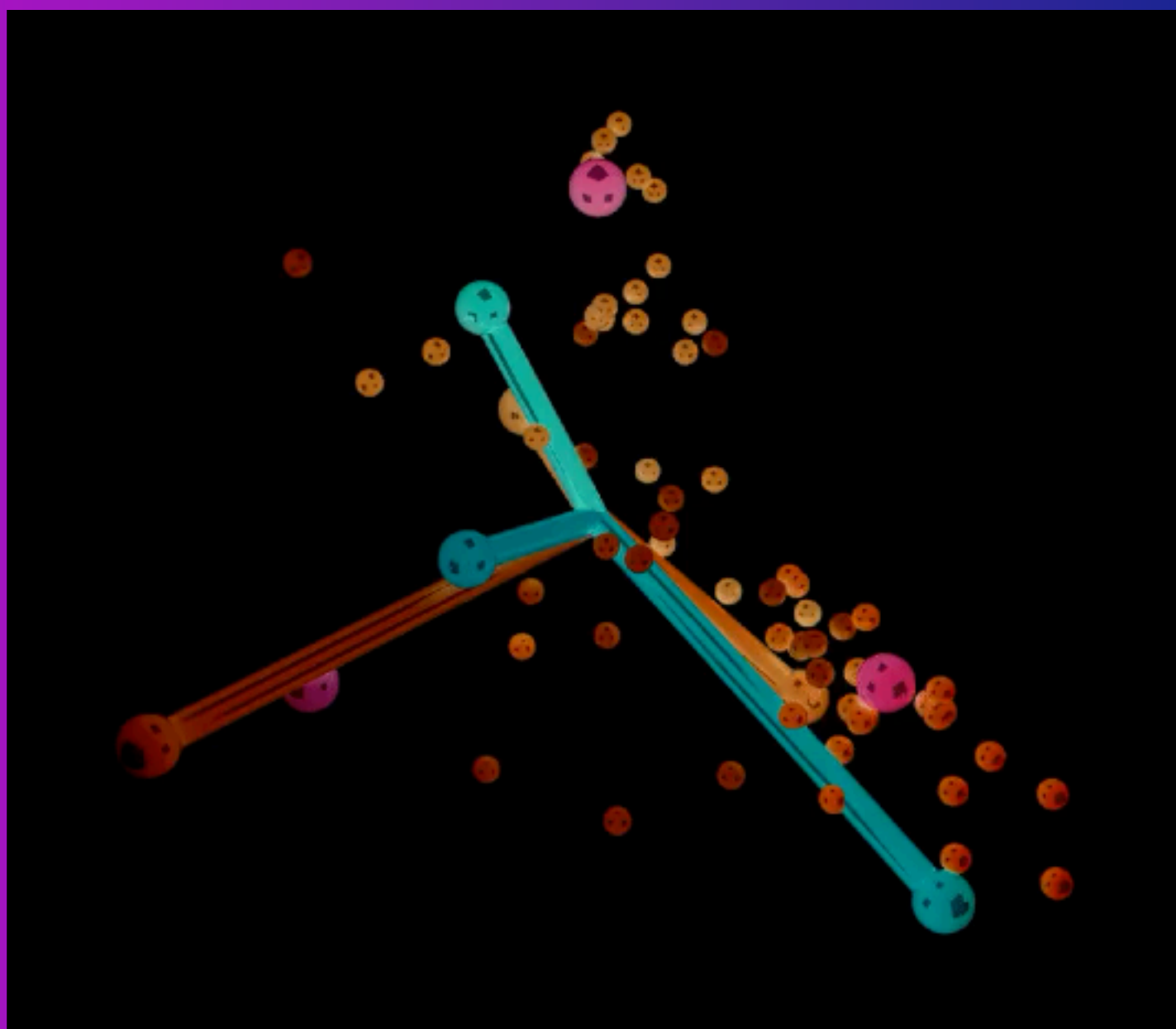


Training Dynamics



Implications & Applications

Neural Collapse

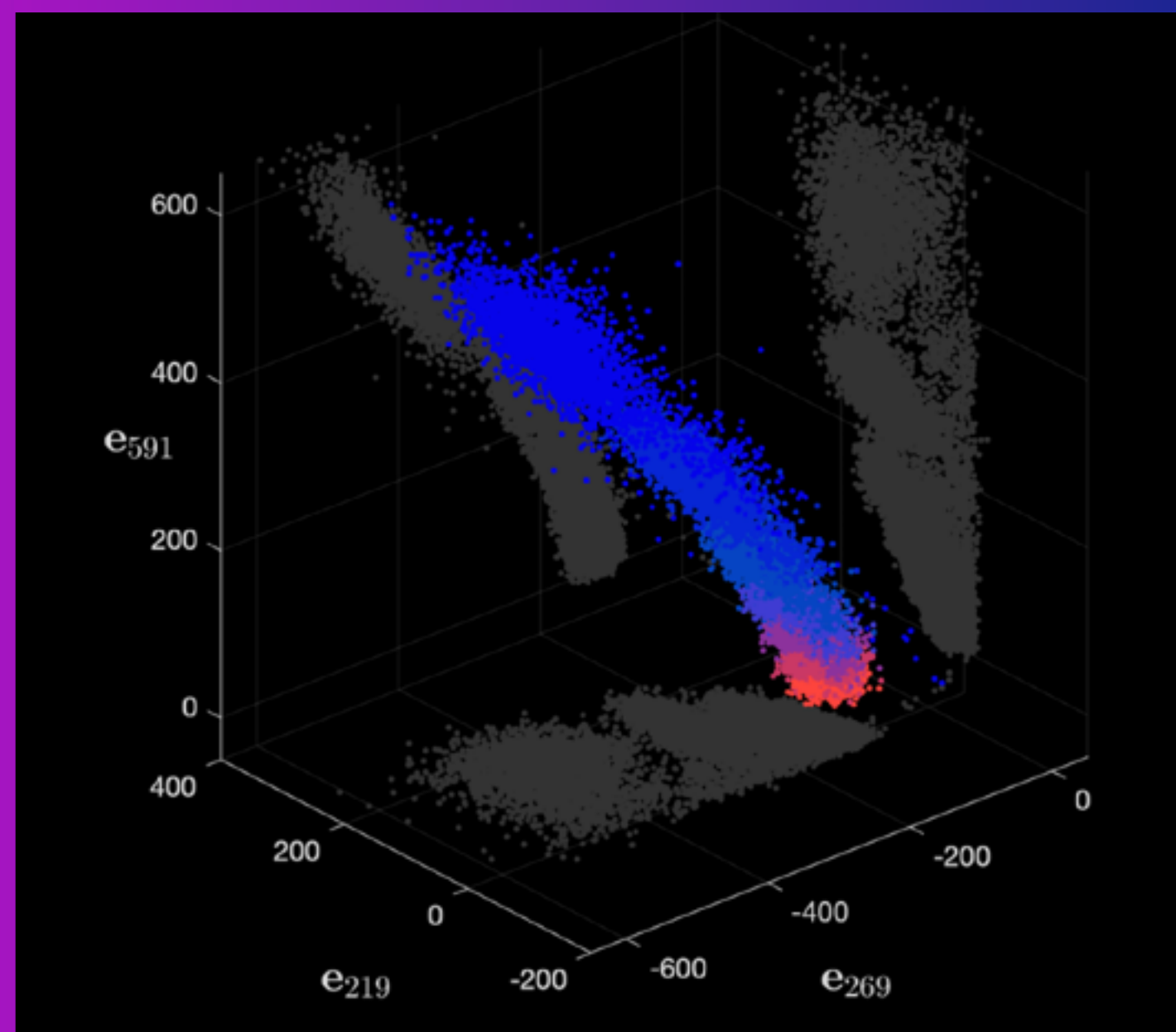


Terminal Phase Separation

- ↪ VGG, ResNet, DenseNet, Transformers
- ↪ Theoretical justification through Average Gradient Outer Product
 - ↪ Beaglehole et. al.

*** work in progress ***

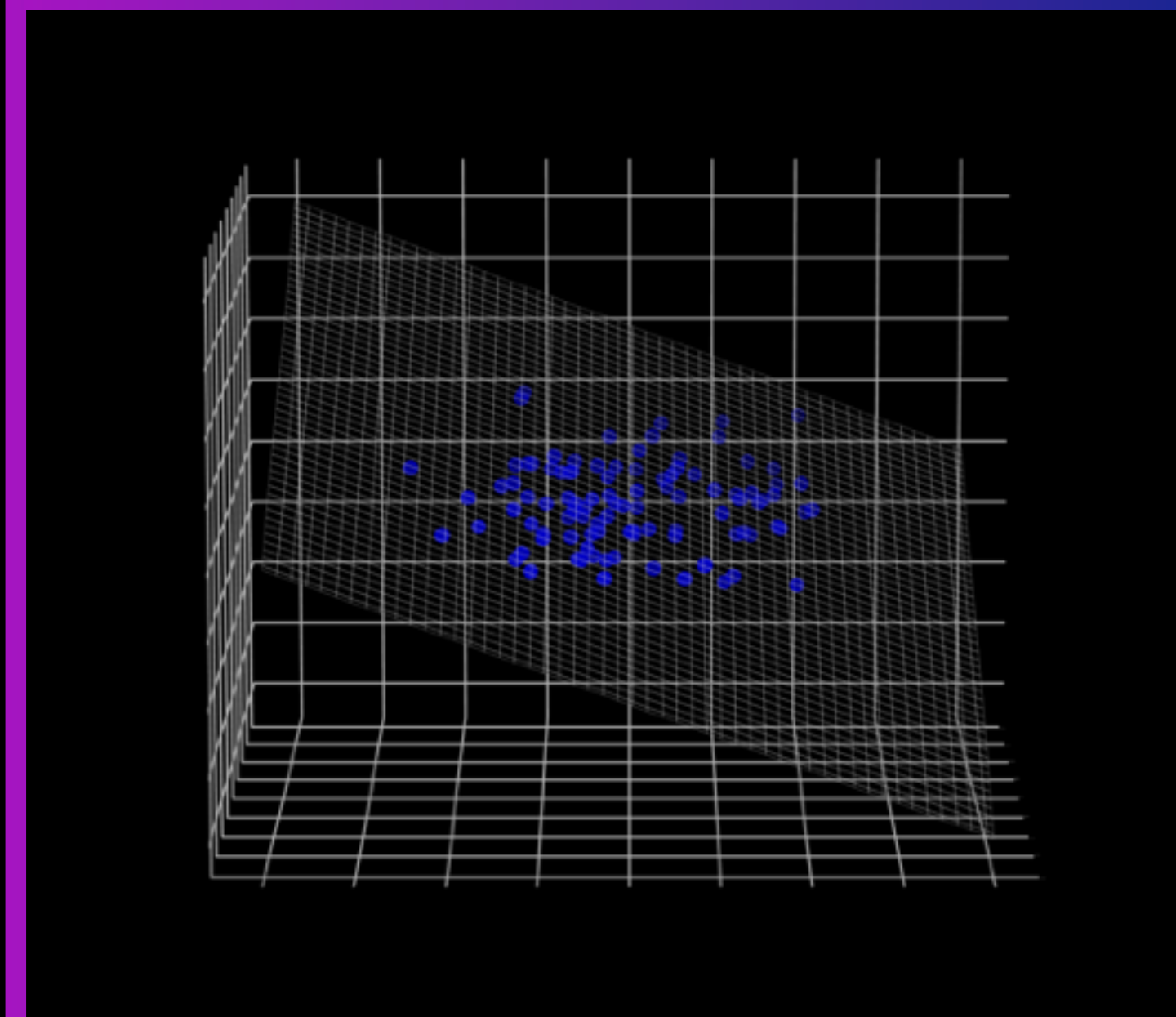
Lines of Thought



Stochastic Trajectory prediction

- ↪ Cornell University; Sarfati et. al.
- ↪ Langevin dynamics for continuous dynamical trajectories
- ↪ Fokker-Plank formulation

Model Compression



Hyperplane Projection

- ↪ Highly targeted model pruning
- ↪ Applications in Edge AI
- ↪ Beyond scope of transformers

*** work in progress ***